

Subject: Cloud ComputingDate: 14/4/25

• Sec-A (2x5=10)

1. Define Virtualization?

Ans Virtualization is the process of creating a virtual version of physical resources such as servers, storage device, networks or operating systems. It allows multiple virtual system to run on a single physical system by abstracting the hardware layer.

2. What are the objectives of virtualization?

Ans Objectives of Virtualization-

- To increase hardware utilization efficiency.
- To reduce IT infrastructure costs.
- To enable flexible and scalable computing.
- To simplify system management and deployment.

3. What are the benefits of storage virtualization in cloud computing?

Ans

- Improve storage utilization through pooling of resources.
- Simplified storage management and provisioning.
- Enhanced data protection and disaster recovery.
- Better scalability and performance.

4. Name two virtualization technologies commonly used in modern data center.

Ans

- VMware vSphere
- Microsoft Hyper-V

5. Explain the concept of improving capacity using virtualization.

Ans Virtualization improves capacity by allowing multiple virtual machines to run on a single physical server. This leads to better utilization of hardware resources, supports dynamic allocation of resources, and enables scaling without adding new physical infrastructure.

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• Sec-B (3x7=21)

1. Describe the role of storage virtualization in modern data management systems. Discuss its significance in enhancing storage flexibility and scalability.

Ans Storage virtualization plays a crucial role in modern data management systems by abstracting physical storage resources and presenting them as a unified, logical storage pool, which can be managed and allocated dynamically. It plays a crucial role in modern data management systems by -

- (i) Abstracting physical storage - Combining multiple storage devices (SAN, NAS, DAS) into a single logical unit, simplifying management.
- (ii) Improving Resource Utilization - Enables efficient allocation of storage capacity across different applications without physical constraints.
- (iii) Supporting Heterogeneous Environment - Allows integration of storage systems from different vendors, enhancing interoperability.
- (iv) Enabling Advanced Data Services - Facilitates features like snapshots, replication, and thin provisioning for better data management.

• Significance of Enhancing Storage Flexibility and Scalability -

(i) Flexibility -

- Dynamic Allocation - Storage can be provisioned or resized on-demand without downtime.
- Migration & Load Balancing - Data can be moved seamlessly across storage systems for optimization.
- Multi-Tenancy Support - Virtualized storage can be partitioned for different users or applications securely.

(ii) Scalability -

- Horizontal & Vertical Expansion - New storage devices can be added without disrupting operations.

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- Elasticity - Storage capacity can scale up or down based on workload requirements.
- Cost-Efficiency - Reduces the need for over-provisioning by allowing just-in-time allocation.

2. Compare and contrast the different approaches to improving performance using virtualization, highlighting the advantages and disadvantages of each method.

Ans Virtualization improves system performance through various techniques, each with distinct advantages and disadvantages. This includes -

(i) Server Virtualization - Multiple virtual machines (VMs) run on a single physical server using a hypervisor.

Advantages:-

- Maximizes hardware utilization
- Reduces hardware and energy costs.
- Easy to scale and manage.

Disadvantages:-

- Performance overhead due to hypervisor
- Resource contention among VMs if not managed properly.

(ii) Storage Virtualization - Combines multiple physical storage devices into a single virtual storage pool.

Advantages -

- simplifies storage management.
- ↳ Improves storage utilization and scalability.
- ↳ Enables data mobility and backup.

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• Disadvantages -

- ↳ Initial Setup Complexity
- ↳ Dependency on virtual storage controllers.

(iii) Network Virtualization - Abstracts physical network resources to create virtual network.

• Advantages -

- ↳ Increase Network flexibility and efficiency.
- ↳ supports automation and fast provisioning.
- ↳ Enhanced security through isolated virtual networks.

• Disadvantages:-

- ↳ complex to configure and maintain.
- ↳ Requires skilled personnel.

(iv) Application Virtualization - Runs applications in isolated containers without installing them on the host systems.

• Advantages:-

- ↳ Reduces compatibility issues.
- ↳ Speeds up deployment.
- ↳ Minimizes system conflicts.

• Disadvantages:-

- ↳ Limited to certain types of applications.
- ↳ May not support all system-level functions.

Q. Explain the concept of improving availability using virtualization.

Discuss the various techniques and strategies employed to enhance system availability in virtualized environments.

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Ans Virtualization enhances system availability by decoupling workloads from physical hardware. This allows systems to continue functioning even during hardware failures or maintenance, ensuring minimal downtime and business continuity.

Techniques and strategies to enhance Availability -

(i) Live Migration - Move running virtual machines (VMs) from one physical host to another without downtime.

• Benefits - Enables maintenance without interrupting services.

(ii) High availability (HA) clustering - Groups of virtual hosts work together to ensure that if one host fails, the VMs automatically restart on another host in the cluster.

• Benefits - Reduces unplanned downtime and ensures quick recovery.

(iii) Fault Tolerance - Creates a real-time mirror of a VM on another host. If the primary fails, the secondary takes over instantly.

• Benefits - Zero downtime, ideal for critical applications.

(iv) Snapshot and Backup Mechanisms - Periodic snapshots or backup of VMs allow restoration in case of data corruption or failure.

Benefits - Fast Recovery and minimal data loss.

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(v) Disaster Recovery with Replication - VMs are replicated to a remote site to ensure business continuity in case of Site-Level Failure.

• Benefits - Protection from natural disaster or major hardware failures.

• Section - c (3x11=33)

1. Discuss the objectives of virtualization in detail; emphasizing its role in improving resource utilization, scalability and flexibility in IT environment. Provide examples to illustrate each objective.

Ans Virtualization is a foundational technology in modern IT infrastructure designed to optimize resource usage, enhance scalability and improve flexibility. Below is the detail discussion -

(i) Improving Resource Utilization -
Objective - To maximize the usage of available hardware by running multiple virtual machines (VMs) on a single physical server.

Techniques & Examples:-

• Server Consolidation - multiple virtual machines run on a single physical server, reducing idle hardware.

eg:- A company replaces 10 underutilized physical servers with 2 high-performance hosts running 20 VMs.

• Dynamic Resource Allocation - Resources are allocated based on demand (eg - VMware DRS, Hyper-V Dynamic Memory).

eg:- A cloud provider automatically assigns more CPU/RAM to a VM during peak loads.

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- Thin Provisioning - Storage is allocated on-demand rather than upfront, reducing waste.

ex:- A VM is given 1TB logical storage but only consumes 200 GB Physically.

- (ii) Enhancing Scalability - ~~To scale~~
Objective - To scale resources up or down based on demand without requiring physical hardware changes.

Explanation - Virtual environments can dynamically allocate CPU, memory and storage to VMs, allowing systems to handle fluctuating workloads efficiently.

ex:- An e-commerce website during peak sales can allocate more resources to its virtualized web servers to handle increased traffic and scale down after the sale.

- (iii) Increasing Flexibility and Agility -
Objective - To enable faster provisioning, deployment and migration of workloads, making IT systems more adaptable.

Explanation - Virtual machines and applications can be deployed or moved quickly across different environments without reconfiguring physical systems.

example:

A development team can quickly spin up isolated test environments using VMs or containers, enabling faster software testing and deployment.

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2. Evaluate the impact of storage virtualization on data management practices considering factors such as cost-effectiveness, data accessibility and data protection. Analyse the challenges associated with implementing storage virtualization and propose strategies to overcome them.

Ans Storage Virtualization abstracts physical storage into a unified Virtual Pool, significantly transforming data management. Below is an evaluation of its impact on cost-effectiveness, data accessibility and data protection, followed by an analysis of implementation challenges and mitigation strategies.

(i) Cost-Effectiveness -

Impact - Storage virtualization consolidates multiple storage systems into a single virtual pool, reducing the need for overprovisioned hardware and improving storage utilization.

• Benefits -

- ↳ Lower hardware and maintenance costs.
- ↳ Reduced energy consumption and physical space.

Ex: A Business can use virtualized storage to manage growing data needs without investing in new physical servers.

(ii) Data Accessibility -

Impact -

Virtual storage allows users and applications to access data from different physical locations as if it were a single system.

• Benefits -

- ↳ Centralized management.

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↳ Faster data access and resource allocation.

Ex: Employees in different branches can access shared files from a single virtual drive, improving collaboration.

(iii) Data Protection -

Impact - Storage virtualization simplifies data replication, backup and recovery.

• Benefits -

↳ Increased data availability.

↳ Easier disaster recovery and business continuity.

Ex: Automatic replication to a backup site ensures data recovery during a system failure.

• Challenges in implementing storage virtualization :-

↳ Complexity in integration with legacy systems.

↳ Initial setup and configuration cost.

↳ Security and data consistency concerns.

↳ Need for skilled IT staff.

• Strategies to Overcome challenges :-

↳ Use vendor-certified solution that integrates easily with existing infrastructure.

↳ Provide training to IT staff and ensure proper documentation.

↳ Implement storage access control, encryption and monitoring tools.

↳ Start with a hybrid model (virtual + physical) to ease the transition.

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Q. Investigate the role of Virtualization technologies in optimizing data storage capacity. Compare and contrast different approaches to capacity improvement, such as thin provisioning, deduplication, and compression. Assess their effectiveness in managing storage resources efficiently.

Ans. Virtualization technologies play a critical role in maximizing storage efficiency by abstracting physical storage and enabling advanced techniques that reduce wastage and improve utilization. Here is the key inside -

(i) Thin provisioning -

Defⁿ - Allocates storage to applications or users on demand rather than assigning all available space upfront.

How it optimizes capacity :-

- ↳ Prevents over-provisioning by assigning storage only when needed.
- ↳ Reduces unused storage space and improves utilization.

• Effectiveness :-

- Highly effective in dynamic environments like cloud or virtual data centers.
- Risk of overcommitment if not monitored properly.

(-ii) Deduplication -

Defⁿ - Eliminates duplicate copies of repeating data blocks, storing only one instance and referencing others to it.

• How it optimizes capacity :-

- ↳ Reduces the volume of redundant data, especially in backup and archival systems.

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↳ Ideal for environments with a lot of similar files (e.g. vms, email systems).

• Effectiveness :-

- ↳ Very effective in backup and file sharing systems.
- ↳ May impact write performance due to additional processing.

(iii) Compression :-

Defⁿ - Reduces the size of stored data by encoding it in a more efficient format.

How it Optimizes Capacity :-

- ↳ Decreases the amount of space required to store files.
- ↳ Works best with text-heavy or compressible data types.

Ex:-

↳ A 10GB log file may be compressed to 2-3 GB without loss.

• Effectiveness :-

- ↳ Effective for archival and infrequently accessed data.
- ↳ May require more CPU resources during read/write.

• Comparison Table -

Techniques	Key Feature	Advantages	Limitations
Thin Provisioning	Allocates storage on demand.	Saves unused space, agile.	Overcommitment risk
Deduplication	Removes duplicate data	Massive space saving in backups.	CPU overhead
Compression	Shrinks data size	Space saving with compressible data	Performance impact for large files.